MATH3202 Operations Research & Mathematical Planning 2026

Assignment 2 – Integer Programming

This assignment is due by 4pm on Thursday, April 30th and is worth 20% of your final grade. You can do this assignment in a group of up to three, with a single submission.

Your job with the Operations Research consulting company is going well. Your boss would like you to continue to work with the Ancestral Skies Collaboration on other aspects of their operations. Communications to you from their team will be provided through Blackboard.

The first communication will appear at 12pm on Thursday, April 2nd with the final communication appearing at 12pm on Thursday, April 22nd.

You will need to prepare a report which includes two main sections:

Section A – Report to your boss

- A general mathematical formulation of the overall problem, including definitions of sets, data, variables, objective function and constraints. *8 marks*
- A Python file with the overall problem modelled for Gurobi. This should be easy to relate back to the formulation. Your boss will attempt to execute this model. *4 marks*

Section B – Report to the client

- Written responses that clearly and concisely address the needs of the client given through each of the communications. *5 marks*
- Brief insights into the solution, such as identifying key constraints or explaining the effects of the additional requirements and options on the score. *3 marks*

Submit your report and Python files via Blackboard, using PDF for the report (saved from Word or created in LaTeX). You must join a group on Blackboard before submitting your assignment but it can be a group of one.

Your Python code should only use Gurobi functions that appear in the examples from lectures and applied classes.

Each student will receive separate data from the client. A group of two or three needs only consider one data set in the report.

Use the report to your boss (Section A) to include an acknowledgement of any AI use in your submission. See the *AI Student Hub* for examples.

Grading Criteria

Section A

Marks	0	1	2
Sets	Incorrect or missing description of sets	Correctly describes sets	
Data	Missing some or all descriptions of data.	Correctly describes most data	Correctly describes all data
Variables	Incorrect or missing description of variables	Correctly describes most variables	Correctly describes all variables
Objective function	Incorrect or missing description of objective function	Correctly describes objective function	
Constraints	Missing many or all descriptions of constraints	Correctly describes most constraints	Correctly describes all constraints.
Python formulation	There is no relationship between Python code and mathematical formulation	Python code mostly matches mathematical formulation	Python code clearly matches mathematical formulation
Python code	Python code does not follow style of class examples	Python code mostly follows style of class examples	Python code follows style of class examples

Section B

Marks	0	1	2	3
Response to communications	Fails to address any of the client questions	Correctly addresses one client question	Correctly addresses three client questions	Correctly addresses all client questions
Written response	Poorly written response with frequent errors in grammar, spelling or technical language; and/or unnecessarily long	Concisely addresses needs of client with few errors in writing	Excellent proficiency in clearly and concisely addressing needs of client	
Insights into the solution	Incorrect or missing insights into solution	Identifies some important factors that affect the solution.	Identifies important factors that affect the solution	Provides insight and thoroughness in identifying factors that affect the solution